StudyMate AI Study Assistant Textbook Test

Chapter 1: Deep Learning Fundamentals

Convolutional Neural Networks (CNNs) use 2D convolution filters to process images.

Pooling layers reduce spatial dimensionality while preserving spatial features.

Fully connected layers compute final class probabilities for prediction.

Chapter 2: Model Training & Optimization

Gradient Descent updates model weights based on loss function gradients.

Adam Optimizer uses adaptive learning rates with momentum.